Is Bohr right? Is it wrong to say that physicists should attempt to figure out how the world actually is? Is it enough to come up with theories that make accurate predictions about the outcomes of experiments? And if Bohm's theory gives the same predictions as "regular" quantum physics (whatever that is), then what's the point of it? How can there be any important difference between two competing theories that make the same predictions?

These questions point to difficult issues in the philosophy of science (some of which we'll meet again in Chapter 8). The short answer is that no, Bohr is not correct, at least not straightforwardly. The picture of the world that comes along with a physical theory is an important component of that theory. Two theories that are identical in their predictions can have wildly different pictures of the world—like putting the Earth at the center of the universe rather than the Sun—and those pictures, in turn, determine a lot about the daily practice of science. If you think that the Sun is at the center of the solar system, rather than the Earth, you're likely to conclude that there's nothing special about Earth, or our solar system, and that there could easily be planets around other stars, even though both astronomical theories give the same predictions about how different lights will move across the sky here on Earth. The story that comes along with a scientific theory influences the experiments that scientists choose to perform, the way new evidence is evaluated, and ultimately guides the search for new theories as well.

In his 1952 papers outlining his new interpretation, Bohm made exactly this point. "The purpose of a theory is not only to correlate the results of observations that we already know how to make," he wrote in the conclusion to his second paper, "but also to suggest the need for new kinds of observations and to predict their results." Bohm laid part of the blame for the Copenhagen interpretation at the feet of logical positivism, the philosophy of science inspired by Mach (which we first encountered in Chapter 3). The Copenhagen interpretation, in Bohm's view, was "guided to a considerable extent" by the idea that objects that can't be seen aren't real, an idea Bohm ascribed to positivism. Yet, as Bohm pointed out, "the history of scientific research is full of examples in which it was very fruitful indeed to assume that certain objects or elements might be real, long before any procedures were known which would permit them to be observed directly."